

Theorem 9.5.1 Computational Formula for $\binom{n}{r}$

The number of subsets of size r (or r -combinations) that can be chosen from a set of n elements, $\binom{n}{r}$, is given by the formula

$$\binom{n}{r} = \frac{P(n, r)}{r!} \quad \text{first version}$$

or, equivalently,

$$\binom{n}{r} = \frac{n!}{r!(n-r)!} \quad \text{second version}$$

where n and r are nonnegative integers with $r \leq n$.